3. Deret Fourier

3.1 Deret Fourier

Jika fungsi f(t) merupakan fungsi periodik dengan periodeT = 2π/ ω, maka deret Fourier dari f(t) berbentuk :

f(t) = a_0 + a_1 cos ωt + a_2 cos 2ωt + … + b_1 cos ωt + b_2 cos 2ωt + ..

$$= a_0 + \sum_{n=1}^{\sim}(a_n \cos n\omega t + b_n \sin n\omega t)$$

f(x) = a_0 + a_1 cos ωt + a_2 cos 2 ωt + a_3 cos 3 ωt + ……+ a_n cos n ωt + …….

+ b_1 sin ωt + b_2 sin 2 ωt + b_3 sin 3 ωt + …… + b_n sin n ωt + …

atau dalam notasi sigma dituliskan sebagai berikut :

$$f(t) = a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega t + b_n \sin n\omega t) \quad (1)$$

Koefisien a_0 , a_n dan b_n ditentukan dengan rumus Euler (Integral) sebagai berikut

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\,dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\,dt \quad (2)$$

Atau

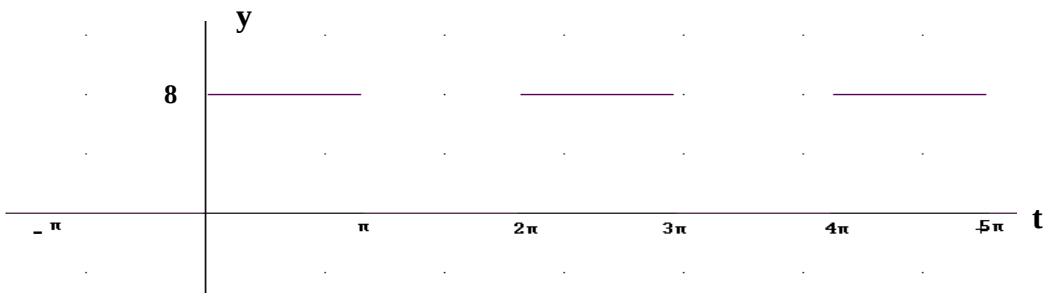
$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos n\omega t\,dt$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\sin n\omega t\,dt \quad (3)$$

Contoh 1. Tentukan deret Fourier fungsi f(t), dan f(t + 2π) = f(t)

$$f(t) = \begin{cases} 0 & \text{Jika } -\pi \le t \le 0 \\ 8 & \text{Jika } 0 \le t \le \pi \end{cases}$$



Gambar 1

Penyelesaian

Periode f(t) adalah T = 2π, atau ω =1.

Koefisien Fourier-nya (rumus 3) adalah

$$\Rightarrow a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,\mathrm{dt} = \frac{1}{2\pi}\int_{-\pi}^{0} 0\,dt + \frac{1}{2\pi}\int_{0}^{\pi} 8dt = \frac{1}{2\pi}(8t\Big|_0^{\pi} = 4$$

$$\Rightarrow a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos \mathrm{nt}\,\mathrm{dt} = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\cos n\,t\,dt \int_{0}^{\pi} 8\cos n\,t\,dt \quad\right]$$

$$= \frac{1}{\pi}\left(\frac{8\sin nt}{n}\Big|_{-\pi}^{0}\right) = \frac{8\sin n\pi}{n\pi} - 0 = 0.$$

$$\Rightarrow b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin \mathrm{nt}\,\mathrm{dt} = \frac{1}{\pi}\int_{0}^{\pi} 8\sin n\,t\,\mathrm{dt}$$

$$= \frac{1}{\pi}\left(\frac{-8\cos nt}{n}\Big|_{-\pi}^{0}\right) = \frac{8}{\pi}\left(\frac{-\cos n\pi}{n} + \frac{\cos 0}{n}\right)$$

$$= \frac{-8}{n\pi}(\cos n\,\pi - 1)$$

Karena cos nπ = -1 untuk n ganjil dan cos nπ = 1 untuk n genap,

$$b_n = \begin{cases} \frac{16}{n\pi} & \text{bila n ganjil} \\ 0 & \text{bila n genap} \end{cases}$$

Koefisien-koefisien b_n adalah

$$b_1 = \frac{16}{\pi}, b_2 = 0, b_3 = \frac{16}{3\pi}, b_4 = 0, b_5 = \frac{16}{5\pi}, b_6 = 0, \ldots\ldots$$

Deret Fourier dari f(t) adalah

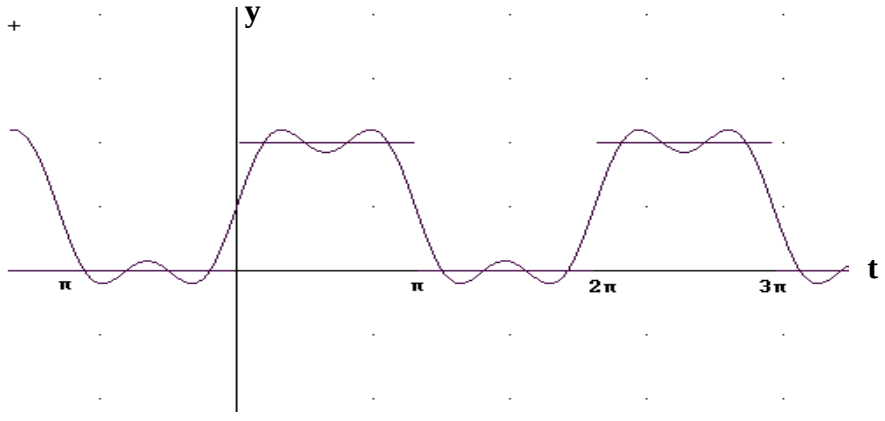
$$f(t) = 1 + \sum_{n=1}^{\infty} b_n \sin n\,t$$

$$= 1 + b_1 \sin t + b_2 \sin 2t + b_3 \sin 3t + b_4 \sin 4t + b_5 \sin 5t + b_6 \sin 6t + \ldots..$$

$$= 1 + \frac{16}{\pi}\sin t + \frac{16}{3\pi}\sin 3\,t + \frac{4}{5\pi}\sin 5\,t + \ldots..$$

Atau $f(t) = 1 + \frac{16}{\pi}\left(\sin t + \frac{1}{3}\sin 3\,t + \frac{1}{5}\sin 5\,t + \ldots..\right)$

Grafik deret Fourier f(t) untuk tiga suku ditunjukkan pada 2.



Gambar 2

Contoh 2

Tentukan deret Fourier fungsi $f(t) = t \quad ; 0 < t < 2\pi \quad dgn\ f(t + 2\pi) = f(t) \quad T = 2\pi.$

Penyelesaian

$$\Rightarrow a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(t)dt = \frac{1}{2\pi}\int_0^{2\pi} t \quad dt = \rightleftarrows \frac{1}{2\pi}\left[\frac{1}{2\pi}t^2\right]_0^{2\pi} \quad = \frac{1}{4\pi}4\pi^2 = \pi$$

$$\Rightarrow a_n = \frac{1}{\pi}\int_0^{2\pi} t\cos n\,t dt = \frac{1}{\pi}\left[\frac{t\sin n\,t}{n} + \frac{\cos n\,t}{n^2}\right]_0^{2\pi} = \frac{1}{\pi n}\left[t\sin n\,t + \frac{\cos n\,t}{n}\right]_0^{2\pi}$$

$$a_n = \frac{1}{\pi n}\left[\left(0 + \frac{\cos 2\pi n}{n}\right) - \left(0 - \frac{1}{n}\right)\right] = \frac{1}{\pi n}\left(\frac{\cos 2\pi n}{n} - \frac{1}{n}\right) = \frac{1}{\pi n^2}(\cos 2\,\pi n - 1) = 0.$$

$$\Rightarrow b_n = \frac{1}{\pi}\int_0^{2\pi} t\sin n\,t dt = \frac{1}{\pi}\left[\frac{-t\cos n\,t}{n} + \frac{\sin n\,t}{n^2}\right]_0^{2\pi} = \frac{1}{n\pi}\left(-tc\cos n\,t + \frac{\sin n\,t}{n}\right)$$

$$b_n = \frac{1}{\pi n}[(-2\pi + 0) - 0] = \frac{-2}{n}$$

Jadi :

$$Deret\ fourier \Rightarrow f(t) = \pi - 2\left(\sin t + \frac{\sin 2\,t}{2} + \frac{\sin 3\,t}{3} + \frac{\sin 4\,t}{4} + \ldots\right)$$

Latihan

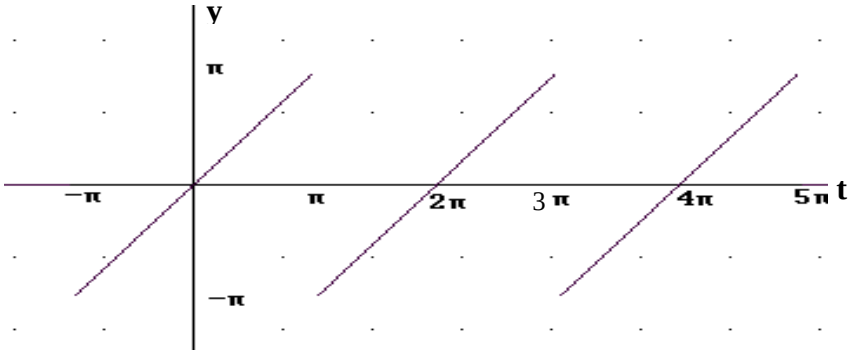
Tentukan deret Fourier fungsi berikut :

1. f(t) = 2,5t ; dengan $0<t<2\pi$, peiode 2π

2. $f(t) = \begin{cases} 1 & \text{bila } -\pi \le t < 0 \\ -1 & \text{bila } 0 \le t < \pi \end{cases}$ dan $f(t + 2\pi) = f(t)$

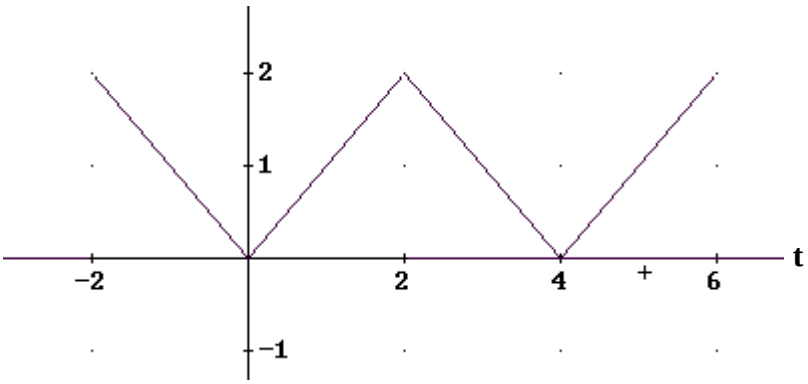
3. $f(t) = \begin{cases} t + 1 & \text{bila } -1 \le t < 0 \\ -t + 1 & \text{bila } 0 \le t < 1 \end{cases}$ dan $f(t + 2) = f(t)$

4. $f(t) = t$ bila $-\pi < t < \pi$, dan $f(t + 2\pi) = f(t)$, lihat gambar berikut :



Gambar .4

5. Gambar 5 yaitu $f(t) = |t|$ bila $-2 < t < 2$ dan $f(t + 4) = f(t)$:



Gambar 5